

Robby Cochran

Technical engineering leader - Security systems, Kubernetes, AI infrastructure, developer platforms

San Francisco, CA | kudzoo@gmail.com | robbycochran.com | github.com/robbycochran | linkedin.com/in/robertacochran

PROFILE

Technical engineering leader with a CS PhD, Staff-level systems/security depth, and people-management experience across startup and enterprise environments. Builds and leads work in technically hard domains: Kubernetes security, eBPF observability, developer platforms, software verification, and AI-agent infrastructure. Open to engineering leadership, Staff+/Principal, forward-deployed engineering, and founder-path roles where product judgment, customer proximity, and deep technical execution all matter.

SELECTED IMPACT

- Built and led runtime-security product work for StackRox/RHACS across behavioral detection, process, network, and file activity, baselines, Kubernetes-native policy, and analyst/developer workflows.
- Helped carry StackRox from early startup product through Red Hat acquisition and into RHACS/OpenShift enterprise scale; product context included \$0 to \$50M+ revenue and 100+ releases.
- Led ACS agentic foundations: internal hackathon, CI tooling, support triage bots, skills strategy, secure harness primitives, and developer paths from local sandbox to OpenShift execution.
- Worked with large customers and design partners to translate deployment constraints into roadmap decisions around scale, compliance, performance, platform support, IBM Power/Z, and false-positive reduction.
- Current focus: runtime contracts, evaluation harnesses, observability, and policy controls for AI agent and developer infrastructure.

EXPERIENCE

IBM / Red Hat

Staff Engineer / Engineering Manager, Red Hat Advanced Cluster Security (RHACS)

2021 - Present

San Francisco, CA / Remote

- Led RHACS runtime-security roadmap spanning process activity, network security, file activity, behavioral baselines, runtime anomaly detection, Kubernetes-native policy workflows, and developer/analyst experience.
- Drove eBPF-based runtime visibility design for OpenShift/Kubernetes environments, including collection semantics, policy exceptions, performance constraints, platform support, and low false-positive detection goals.
- Built and guided secure AI-agent infrastructure through harness-openshell: declarative workflows, provider/credential boundaries, inference routing, sandbox execution, and repeatable CI validation.
- Managed distributed engineering teams while staying close to architecture and implementation; promoted L4 and L5 engineers through scope design, feedback, and career planning.
- Shaped OpenShift-first RHACS strategy around console integration, RBAC alignment, GitOps/policy-as-code, runtime observability, and AI-assisted workflows. Public talk: eBPF 101.

StackRox, Inc. (acquired by Red Hat)

Staff Software Engineer / Senior Software Engineer / Member of Technical Staff

2017 - 2021

Mountain View, CA

- Joined an early-stage Kubernetes security startup and built runtime detection and enforcement features for containerized workloads and Kubernetes clusters.
- Advanced core runtime-security capabilities including baseline-driven detection, process and network visibility, and connections between observed runtime behavior and security policy.
- Worked across product, engineering, and customer-driven security cases with large design partners, helping mature the platform through acquisition by Red Hat.
- Patent: US 11,811,804, "System and method for detecting process anomalies in a distributed computation system utilizing containers."
- Talk: BSidesSF 2018 - Listen to your Engine.

University of North Carolina at Chapel Hill

PhD Researcher, Computer Security and Software Verification

2008 - 2016

Chapel Hill, NC

- Developed symbolic client verification: server-side techniques for checking whether observed client network traffic could have been produced by sanctioned software.
- Built symbolic-execution systems for online games, cryptographic protocols, and distributed applications; refactored KLEE internals for parallelism and performance research.
- Publications: NDSS 2010 Best Paper / TISSEC 2011 on online-game client verification; NDSS 2013 on distributed-application client behavior; NSDI 2017 on known cryptographic clients.

Microsoft Research - Research in Software Engineering (RiSE)

Research Intern

Summer 2013

Redmond, WA

- Program synthesis via crowdsourcing: combined symbolic automata, genetic programming, and online workers; publication: POPL 2015 "Program Boosting"; patent: US 9,753,696.

Intel Corporation - Visual Computing Group

Software Engineer

2006 - 2007

Hillsboro, OR

- Built graphics-hardware analysis tooling to interpret, record, and replay Direct3D/OpenGL API calls for performance analysis of prototype visual-computing hardware.

EDUCATION

PhD, Computer Science, University of North Carolina at Chapel Hill, 2016 - Advisor: Michael K. Reiter. **BS, Computer Science**, Clemson University, Magna Cum Laude, 2006.

TECHNICAL DOMAINS

Runtime security; Kubernetes; OpenShift; RHACS; eBPF observability; behavioral baselines; policy-as-code; distributed systems; symbolic execution; software verification; formal methods; program synthesis; AI-agent safety; LLM gateways; MCP; Go; C/C++; Python; Rust; Bash; Docker; LLVM; SMT/SAT solvers.